PART 3 — INTRODUCTION

Visual-Omics Foundation Models

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EXERCISE

Hands-on — open the notebook

```
notebooks/Part3_OmiCLIP_Introduction.ipynb
```

In the next part you will:

1. Load the pretrained OmiCLIP model from the Loki package.
2. Encode ~1 000 Visium spots from a breast-cancer slide.
3. Inspect the cosine-similarity matrix and a UMAP of the joint space.
4. Run image → transcriptomics retrieval and visualize the top-k matches.

RECAP

Part 3 — Section summary

OmiCLIP = CLIP with a transcriptome on the “text” side

Vision-transformer (H&E) + expression-transformer (rank-ordered genes), contrastive InfoNCE.

Pretrained on ~2 M Visium spots, ~100 datasets

Diverse tissues, vendors, chemistries → broad transfer to downstream tasks.

The shared latent space is the primitive used by Loki & Thor in Part 4

Alignment, annotation, decomposition, retrieval, and H&E→expression all reduce to operations on this space.

P A R T 4 — A P P L I C A T I O N # 1

From foundation model to practical analysis

Loki — A platform for visual–omics foundation model OmiCLIP
Thor — A cell-level investigation of spatial transcriptomics and histology platform

Guangyu Wang's group · Houston Methodist Research Institute

github.com/GuangyuWangLab2021/Loki · github.com/GuangyuWangLab2021/Thor

EXERCISE

Hands-on — Loki & Thor in practice

```
notebooks/Part4_Loki_Thor_Applications.ipynb
```

In the next part you will:

1. Align two consecutive Visium sections with Loki and inspect the alignment error.
2. Annotate spots with a TCGA bulk reference and with a custom marker panel.
3. Decompose spots into cell-type proportions using a single-cell breast atlas.
4. Retrieve transcriptomes from an H&E region of interest.
5. Run Thor on a TCGA H&E slide; correlate predicted ERBB2 with clinical HER2 status.

PACKAGE

Loki at a glance

`Open-source Python package — github.com/GuangyuWangLab2021/Loki`

Five core modules. Each takes scanpy/AnnData in and returns AnnData out — minimal friction with existing pipelines.

`loki.align`	Visium ↔ Visium tissue alignment via OmiCLIP image features.
`loki.annotate`	Bulk-RNA-seq and marker-gene annotation in the shared latent space.
`loki.decompose`	Cell-type decomposition by NNLS on reference centroids.
`loki.retrieve`	Image ↔ transcriptomics retrieval (per-spot and slide-scale).
`loki.models`	Convenience loaders for OmiCLIP checkpoints.

Loki #1 — Tissue alignment

- Problem — consecutive Visium sections don't share spot coordinates: rotation, translation, deformation, missing tissue.
- Use OmiCLIP image embeddings as a modality-invariant feature space; match spots by cosine similarity.
- Fit affine / thin-plate-spline warp. Optional MNN refinement.
- Three feature modes — image (robust across patients), expression (within-patient), joint.

Mouse-brain alignment benchmark

Method	Mean err. (µm)
PASTE	79
STalign	51
Loki (image)	**38**

Use 'tps' for tumours with significant deformation.

```
from loki.align import align_sections
result = align_sections(secA, secB, encoder=model, feature='image', transform='affine', refine_with_mnn=True)
```

Loki #2 — Annotation by bulk RNA-seq

- Input: any bulk RNA-seq reference matrix (TCGA subtypes, GTEx tissues, in-house).
- Push every reference column through the expression encoder once — get a 512-d reference per label.
- Assign each Visium spot to the closest reference by cosine similarity.
- Gene overlap doesn't need to be 1-to-1 — only the gene tokens matter.
- No retraining, no per-dataset hyper-parameter tuning.

Bulk reference

TCGA-BRCA Luminal A / B / HER2+ / Basal

↓

OmiCLIP expression encoder

→ 512-d embedding per subtype

↓

Per-spot label

argmax cosine similarity
+ confidence score

Loki #3 — Annotation by marker-gene panel

- When no bulk reference is available, supply a dictionary { cell-type → [marker genes] }.
- Each list is converted to a 'pseudo-expression' token sequence (uniform expression on the markers).
- Encode → reference embedding per panel → nearest match per spot.
- Works with as few as 3–5 markers — great for exploratory analysis on new tissues.

```
marker_panels = {
    'Tumor epithelium' : ['ERBB2', 'EPCAM', 'KRT8', 'KRT18', 'CDH1'],
    'T cells'          : ['CD3D', 'CD3E', 'CD8A', 'CD4', 'GZMB'],
    'Stromal/CAF'      : ['COL1A1', 'COL3A1', 'FAP', 'ACTA2', 'PDGFRB'],
    'Endothelium'      : ['PECAM1', 'VWF', 'CDH5', 'CLDN5'],
    'Macrophages'      : ['CD68', 'CD163', 'C1QA', 'C1QB', 'AIF1'],
}
annot = marker_annotate(adata, marker_panels, encoder=model, tokenize=tokenize_expression)
```

Loki #4 — Cell-type decomposition

- A Visium spot ≈ a mixture of 1–20 cells.
- Reference embeddings come from any annotated single-cell atlas.
- Non-negative least squares in 512-d (closed-form, < 1 ms / spot).
- Returns per-spot cell-type proportions → standard sc.pl.spatial.
- Faster and more robust than gene-space methods because we work in a denoised low-dimensional space.

Loki #5 — Cross-modal retrieval

- Build an index of Visium spots → expression embeddings.
- Query with any H&E patch — Visium or not (any pathology slide).
- Return nearest spots and their gene-level expression profiles.
- Use case: pathologist circles a region of interest on a slide; system retrieves matching transcriptomic profiles from a curated atlas, with top upregulated genes.

Turns any tissue atlas into a queryable resource for everyday pathology work.

Thor — predicting gene expression from H&E alone

1. Tile H&E slide

224 × 224 px patches, configurable stride

2. Encode tiles

OmiCLIP image encoder (frozen)

3. Predict expression

Linear or 2-layer MLP head

4. Stitch map

Per-gene spatial expression map

The regression head is trained once on ~2 M Visium spots and then applied to any H&E slide.

⇒ Bring molecular maps to TCGA, biobank cohorts, and routine pathology archives — slides where Visium was never run.

Thor — what works, what doesn't

✓ Works well

- Highly expressed lineage / compartment genes (KRT8, CD3D, COL1A1, ERBB2, …).
- Hallmark / pathway-level signals.
- Spatial domains with clear morphological correlates.
- Per-gene Pearson r ≈ 0.4–0.7 for the top decile of HVGs.

✕ Be careful

- Lowly expressed TFs, ligands, cytokines — high noise.
- Rare programs without obvious morphological correlate.
- Extreme stain variability (use Macenko-style normalization first).
- Per-gene r near zero in the lowest decile — always inspect the gene list.

CASE STUDY

End-to-end case — TCGA-BRCA H&E slide

TCGA H&E (no Visium) → Thor prediction → Predicted expression map → Loki annotation → Maps + stats

Outputs we get for free

- Per-gene spatial expression maps (ERBB2, ESR1, MKI67, CD3D, COL1A1, …).
- Inferred tissue compartments (tumour / stroma / immune / vasculature) overlaid on the slide.
- Surrogate HER2 / ER / PR scoring correlated with the clinical metadata table.
- Slide-level summary embedding for cohort-scale comparisons.

TIPS

Practical advice

Always inspect a UMAP first

Colour by modality. If image and expression embeddings are separated, something is off upstream.

For decomposition, invest in the reference

The reference single-cell atlas matters more than the decomposition method.

For Thor, validate on held-out Visium

Compute per-gene Pearson r; trust the top decile, treat the bottom decile as noise.

Start small

The base OmiCLIP checkpoint is enough for most tasks. Embeddings are cheap to compute.

GET STARTED

Resources

Loki	github.com/GuangyuWangLab2021/Loki
Thor	github.com/GuangyuWangLab2021/Thor

Q & A

Thank you!

github.com/GuangyuWangLab2021